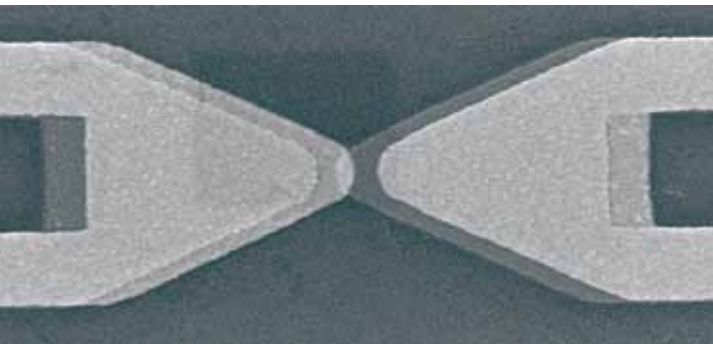


Rectifying antennas that harvest power from heat

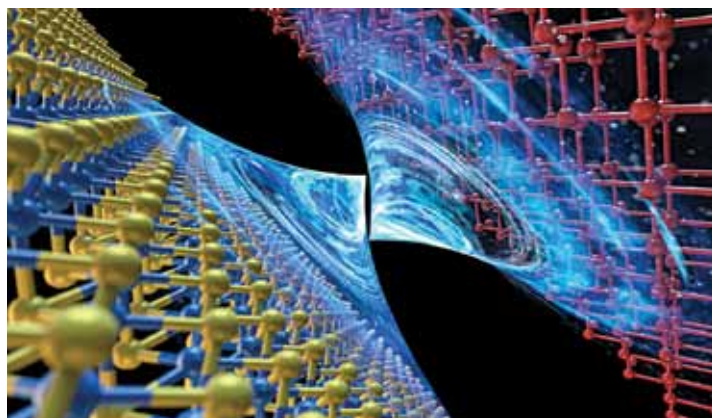


This scanning electron microscope image shows the distinct bow tie shape of an optical rectenna (Credit: www.colorado.edu)

Scientists at CU Boulder have tapped into a property of electrons to design devices that can capture excess heat from their environment and convert it into usable electricity. The devices known as ‘rectennas’ (rectifying antennas) are too small to see with the naked eye and are roughly 100 times more efficient than similar tools used for energy harvesting. This is achieved through a process called ‘resonant tunnelling’ in which electrons pass through solid matter without spending any energy. Rectennas work a bit like car radio antennas. But instead of picking up radio waves and turning them into tunes, optical rectennas absorb light and heat and convert it into power. Working rectennas could, theoretically, harvest the heat coming from factory smokestacks or bakery ovens that would otherwise go waste.

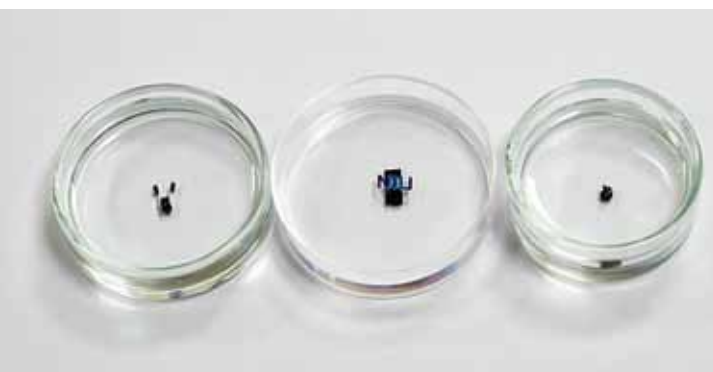
Atomically thin materials instead of silicon as the basis for transistors

The use of atomically thin materials instead of silicon as the basis for new transistors is exciting but connecting those ‘2D’ materials to other conventional electronic components has proved difficult. Now researchers at MIT, the University of California at Berkeley, the Taiwan Semiconductor Manufacturing Company, and elsewhere have found a new way of making those electrical connections, which could help to unleash the potential of 2D materials and further the miniaturisation of components—possibly enough to extend Moore’s Law, at least for the near future. The 2D materials or sheets, just a few atoms thick, meet all the requirements for enabling miniaturisation, potentially reducing by several times a key parameter called the channel length, from around 5 to 10 nanometres in current cutting-edge chips to a sub-nanometer scale.



At the interface between the semimetal (bismuth) and the 2D semiconductor (MoS2), there is no energy barrier for the electron to go through, leading to an ultralow contact resistance between them (Credit: <https://news.mit.edu>)

Manoeuvrable miniature robots controlled by magnetic field



Millimetre-sized robots (Credit: www.ntu.edu.sg)

A team of scientists at NTU Singapore has developed millimetre-size robots that can be controlled using magnetic fields to perform highly manoeuvrable and dexterous manipulations. This could pave the way to possible future applications in biomedicine and manufacturing. The research team created the miniature robots by embedding magnetic microparticles into biocompatible polymers—non-toxic materials that are harmless to humans. The robots are ‘programmed’ to execute their desired functionalities when magnetic fields are applied. The robots improve many existing small-scale robots by optimising their ability to move in six degrees of freedom (DoF)—that is, translational movement along the three spatial axes, and rotational movement about those three axes, commonly known as roll, pitch, and yaw angles.